24/12/24 MS

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[22ECES502]

FIFTH SEMESTER B.Tech EXAMINATION, DECEMBER – 2024

IoT IN AUTOMOTIVE SYSTEMS

Time: 3 Hours **Maximum Marks: 100**

Instructions: i. Missing data may be suitably assumed.

ANSWER ALL QUESTIONS

		PART – A	**5 x 2 = 10**		
1.	**a.**	What is the role of the middleware layer in IoT architecture?	L1	CO1	**02**
	b.	What is the primary function of servers in an IoT ecosystem?	L1	CO2	**02**
	c.	What is the purpose of the ADC in an Arduino board?	L1	CO3	**02**
	d.	What are the steps involved in interfacing ESP8266 with an analog sensor?	L1	CO4	**02**
	e.	Describe the role of the ThingSpeak server in the engine oil and coolant level monitoring system.	L2	CO6	**02**
		PART – B	**6 x 5 = 30**		
2.	**a.**	What is an IoT communication API? Explain its role in IoT development and give an example of a commonly used IoT API.	L2	CO1	**05**
		OR			
	b.	Explain in detail any three key technologies used in Industry 4.0 and explain their roles in transforming manufacturing processes.	L2	CO1	**05**
3.	**a.**	Explain in detail the role of IoT servers in managing devices and data within an IoT ecosystem, focusing on any four platforms.	L2	CO2	**05**
		OR			
	b.	Outline the key steps involved in the design methodology of IoT devices.	L2	CO2	**05**
4.	**a.**	Explain serial communication between the ESP8266 and PC with a neat diagram.	L2	CO4	**05**
		OR			
	b.	Discuss the key steps involved in establishing connection to a Wi-Fi network for NodeMCU microcontroller.	L2	CO4	**05**
5.	**a.**	Explain the process of interfacing Blynk app with NodeMCU microcontroller.	L2	CO4	**05**
		OR			
	b.	Compare the advantages and disadvantages of analog and digital sensors with the ESP8266 in IoT applications, which one is more advantageous.	L2	CO4	**05**
6.	**a.**	Discuss the advantages of using ThingSpeak for real-time monitoring in the engine Oil and coolant level monitoring system.	L3	CO5	**05**
		OR			
	b.	Describe the role of IoT integration in enhancing the efficiency of the Fleet and Driver Management System.	L2	CO5	**05**

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7.	a.	Describe the role of each component involved in the real-time car telematics tracking system with necessary diagram.	L2	CO6	05
		OR			
	b.	Discuss the role of IoT in the talking road unit and explain how it enhances road safety for drivers.	L2	CO6	05
		PART – C	**6 x 10 = 60**		
8.	a.	Describe the IoT reference model. Explain its layers and the purpose of each layer.	L2	CO1	10
		OR			
	b.	Elaborate the different IoT communication models and discuss how these models facilitate data exchange in IoT systems.	L2	CO1	10
9.	a.	Describe any four levels of IoT with a neat diagram and relevant examples.	L2	CO1	10
		OR			
	b.	Explain the role of GPRS in IoT communication and how it interacts with a microcontroller for data transmission.	L2	CO2	10
10.	a.	Explain the process of serial communication between an Arduino and RF modem with necessary diagram and program code.	L2	CO3	10
		OR			
	b.	Explain the interfacing process of a DC voltage sensor and an LCD display with an Arduino with circuit diagram and program code.	L2	CO3	10
11.	a.	Write a program code for the biometric car door opening system using the ESP8266 and include necessary libraries.	L2	CO4	10
		OR			
	b.	Describe the process of interfacing NodeMCU with an Arduino microcontroller for an accident monitoring System with block diagram and pseudocode.	L2	CO4	10
12.	a.	Explain the working of an engine oil and coolant level monitoring system with block diagram, and discuss the role of ThingSpeak server.	L2	CO5	10
		OR			
	b.	Write a simplified pseudocode for the fleet and driver management system and describe its key functions.	L2	CO5	10
13.	a.	What is the role of the ThingSpeak server in a real-time car telematics tracking system, and how does it handle data?	L2	CO6	10
		OR			
	b.	Explain the components and block diagram of a talking road unit designed for pin turns in hilly areas.	L2	CO6	10